

Chris Dietrich

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TECHNICAL SKILLS

Languages: C++, Python, C# , Java, SQL

Tools: GitHub, AutoCAD, SolidWorks, Xcelgo Experior

Technologies/Frameworks: Unity, OneDrive, Excel, REST API, ROS (learning), OpenCV (learning)

EDUCATION

Candidate for Bachelor of Applied Science University of Waterloo

2025 – Present

Bachelor's of Applied Science in Mechatronics Engineering.

AFNORTH International Highschool Ontario Diploma (98.7% graduating average)

2021 – 2025

WORK EXPERIENCE

Software Engineer Brock Solutions

Jan-Apr 2026

- Built a C# tool enabling remote interaction with Experior (Xcelgo) airport baggage handling system simulations and emulations, allowing multiple engineers to access shared models and reducing licensing costs.
- Implemented REST API messaging between C# WinForms frontend and Experior C# backend models.
- Developed a data synthesizing application using Microsoft SQL Client to query airport operational databases, call stored procedures, and synthesize real-time data into XML/CSV for simulation inputs, reducing data preparation time from 6 hours to 5 minutes.
- Created an AI-assisted C# GUI development workflow, doubling frontend productivity.
- Set up and networked Virtual Machines to implement airport baggage tracking software.

PROJECTS

Python projects

2023 – Present

Designer

- Designed a Python script for flash card studying of trombone slide positions, enabling me to learn Python whilst trombone.
- Developed a calculator to factor, convert to vertex form, and find roots of a given function to simplify math homework.
- Developed a script to simulate keyboard inputs in a video game to autonomously generate game resources.

Unity game development

2024 – 2025

Developer

- Developed two games in unity placing first at the Future Business Leaders of America provincial competition
- Utilized a Unity git repository for active collaboration with a Co-developer

Voice assistant Developer	2025 – Present
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- Developed working Speech To Text (STT) and Text To Speech (TTS) system and custom wake word.
- Developing a working voice assistant in Python to do arithmetic, simple computer functions and search queries.
- Currently coding intent recognition on incoming STT queries.

Autonomous Tic Tac Toe Robot Class Project	2025 – Present
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- Developed a fully autonomous robot to play Tic Tac Toe against a human player, using VEX IQ as the control hub, for the MTE100 class.
- Used a shared GIT repository to keep our team's code in synch.

EXTRACURRICULARS

Future Business Leaders of America (FBLA) club Club President	2023 – 2025
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- Led 30 members in fundraising efforts, collecting over \$2500 for student initiatives

FIRST Robotics Team Team Lead	2022 – 2025
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- Led a team of six in developing a robot to compete in the FIRST Tech Challenge.
- Developed several core components for the robots such as a grappling hook, angled slide, and claw designs winning our team the “Think” and “Innovate” awards.
- Developed Java code for autonomous robot navigation, increasing average autonomous phase score by 20 points per match.
- Won FIRST Tech scholarship valued at \$15000.

Waterloo Aerial Robotics Group (WARG) Member	2025 – Present
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- Currently enrolled member of WARG Autonomy sub team planning to specialize in computer vision this term

AWARDS

Schulich Leaders Scholarship	2025
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- Selected as school nominee, for leadership and creativity.

Valedictorian	2025
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- Honored for academic excellence (98.7% average) and leadership.

Graf Von Kielmannsegg	2025
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- Recognized by *NATO Base commander General Guglielmo Luigi Miglietta* for academic excellence, community contribution, leadership and character.

Future Business Leaders of America	2023 - 2025
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- First place awards for video game development using Unity X3, marketing, economics, coding and programming, international business, financial math, business management, and business education.